

PHYLUM ARTHROPODA

Introduction

The significance of this lesson is that *it presents you with a chance to learn about the largest and most diverse phylum in the animal kingdom.*

In addition, arthropods and vertebrates are the only two animal groups with many members fully adapted for life on land.

General characteristics and adaptive features of arthropods

- Arthropoda means jointed legs (arthro=joint; poda=legs/feet).
- Arthropods and vertebrates are the only two animal groups with many members fully adapted to life on land. The oldest known metazoan (*having a body made up of differentiated cells arranged in tissues and organs.*) animal fossils are arthropods.
- These were centipede and spider-like forms dating from 414 million years ago (Silurian period).
- About 75% (over 1 million species of all animal species described to date belong to the phylum Arthropoda making it the most successful in the animal kingdom with the **class insecta accounting for the most majority.**
- It is speculated that, there are many more species of arthropods especially in the tropics that are yet to be discovered.
- Arthropods occupy about every conceivable ecologic niche from low ocean depths to very high altitudes and from tropics far into both north and south polar regions.
- They occur in the sea, in fresh water, brackish water, on land, in air and the bodies of other animals.
- Members of this phylum vary enormously in size, shape and habits.
- They range from microscopic mites to giant Japanese spider crab (10-12ft).
- Arthropods are thought to have arisen from an annelid-like ancestor(s) because the basic body plan is a **long-segmented body.**
- Trilobites are fossil arthropod forms which had this type of body form (**trilobites extinct marine arthropods that form the class Trilobita. Trilobites**

form one of the earliest known groups of arthropods.). This ancestral body plan had **one pair of jointed appendages per segment**, as in modern centipedes.

- The annelidan ancestry of arthropods is further reflected in their **segmentation, the plan of their nervous system, and their determinate cleavage**.
- They are bilaterally symmetrical, triploblastic coelomate metazoans that operate at organ system level.
- They show metameric segmentation, with **tagmosis** (*the evolutionary process by which segments in an animal's body combine to form distinct, functional units called tagmata*) leading to formation of various body parts or tagmata e.g head, thorax and abdomen.
- They possess a **chitinous** (*a fibrous substance consisting of [polysaccharides](#), which is the major constituent in the [exoskeleton](#) of [arthropods](#)*) exoskeleton and in order to conserve water the outer cuticle (epicuticle) is waxy.
- Growth and development proceeds via a series of moults (*the process by which insects shed their exoskeleton, or cuticle, to grow a new one. It's also known as ecdysis.*) and intermolt growth periods.
- They have a complete digestive system with both a mouth and anus.
- They possess **striated muscles** (*muscle tissue that is marked by transverse dark and light bands, is made up of elongated usually multinucleated fibers, and includes skeletal muscle, cardiac muscle, and most muscle of arthropods*). *Striated muscle has a much faster contraction rate than smooth muscle, and it is this feature that probably enabled the development of flight in many insects.*
- Their central nervous system has an anterior supra oesophageal ganglion ("brain") (*a part of an insect's central nervous system that controls the mouthparts, neck muscles, and salivary glands*) connected to the ventral nerve cord by connective nerves.
- Their coelom is generally reduced and is filled with blood (haemolymph) and is called haemocoel.
- A dorsal heart (dorsal vessel) with **ostia** (*small, paired openings in the heart of insects that allow hemolymph to flow into and out of the heart*) is present.
- The sexes are typically **separate** and fertilization is mainly internal except for minute aquatic spp who practice external fertilization.

- They have variously modified jointed appendages which may take the form of antennae, mouthparts (mandibles, maxilla, labrum, labium), external genitalia (ovipositor, claspers), legs, claws, anal cerci.
- They have variously modified sense organs which include compound eyes, simple eyes, antennae, anal cerci, mandibular palps, maxillary palp, labial palps, pads on their feet and the numerous bristles found on their legs.
- To further conserve water, most excrete solid nitrogenous waste products in form of **uric acid**.
- Their blood pigment is called **haemocyanin** which contains copper that become blue on oxidation after exposure to air.

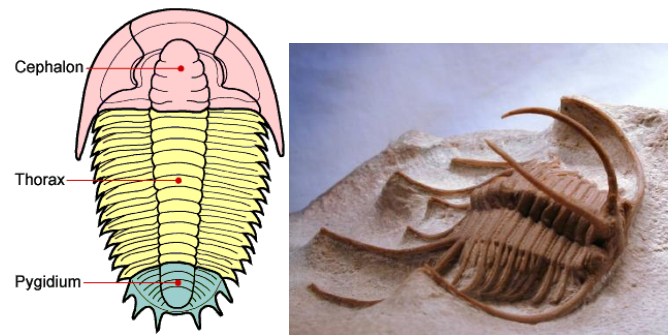
Classification

The phylum Arthropoda is made up of five classes namely:

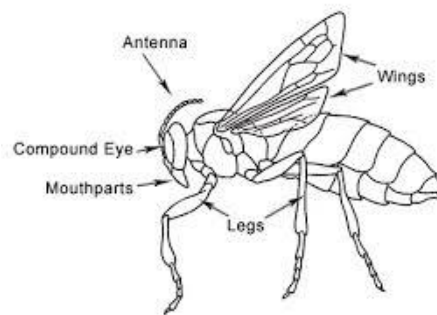
- ❖ Trilobita (extinct)
- ❖ Arachnida,
- ❖ Crustacea,
- ❖ Insecta
- ❖ Myriapoda

Class	Examples	Body Segmentation	Appendages	Respiration	Distinctive Features
Trilobita (<i>extinct</i>)	Trilobites (fossil forms)	Three-lobed, three-tagmata (head, thorax, pygidium)	Biramous appendages	Gills (branchial structures)	Earliest arthropods; extinct since Permian period
Arachnida	Spiders, scorpions, ticks, mites	Two tagmata (cephalothorax + abdomen)	4 pairs of walking legs; chelicerae, pedipalps	Book lungs or tracheae	No antennae; specialized for terrestrial life
Crustacea	Crabs, lobsters, shrimps, barnacles	Head, thorax, abdomen (often fused into cephalothorax)	Biramous appendages; 2 pairs of antennae	Gills	Mostly aquatic; exoskeleton often calcified
Insecta	Beetles, butterflies, ants, flies	Three tagmata (head, thorax, abdomen)	3 pairs of legs; usually 2 pairs of wings; 1 pair antennae	Tracheal system	Largest class; adapted to diverse habitats
Myriapoda	Centipedes, millipedes	Head + long segmented trunk	Uniramous appendages; many legs (1 pair per segment in centipedes, 2 in millipedes)	Tracheae	Elongated bodies; terrestrial detritivores or predators

Class Trilobita (Extinct)



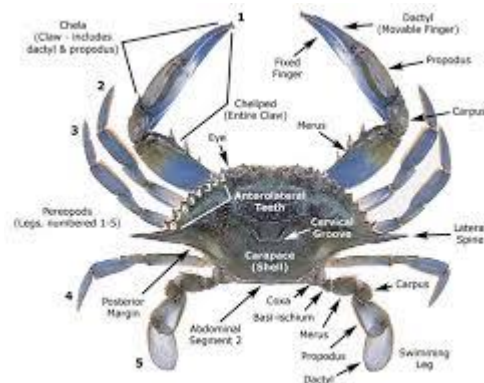
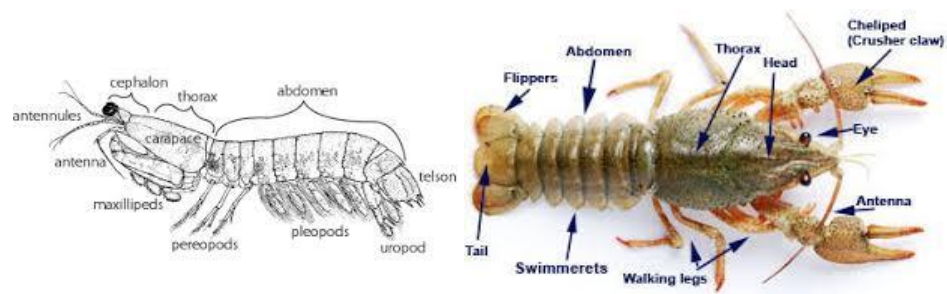
Class Insecta



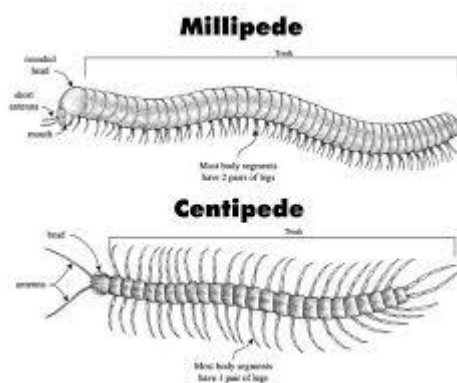
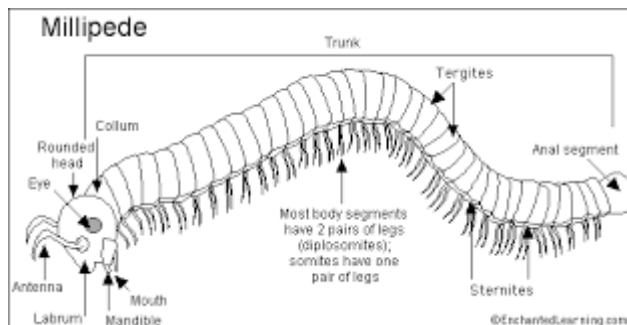
Class Arachnida



Class Crustacea



Class Myriapoda



FACTORS THAT HAVE CONTRIBUTED TO THE SUCCESS OF ARTHROPODS

- Arthropods are the most successful animals on Earth because of their protective exoskeleton, segmented body with specialized appendages, advanced sensory systems, efficient respiratory system, high reproductive potential, diverse feeding strategies, and ecological adaptability.

These traits allow them to thrive in nearly every habitat, including Kenya's savannahs, farms, and coastal ecosystems. These traits explain why they make up over 80% of all known animal species worldwide and remain dominant in ecosystems, including those in Kenya.

➤ **Structural Adaptations**

- **Exoskeleton of chitin:** Provides armor against predators, prevents water loss (critical in arid regions), and offers strong muscle attachment which provides a rigid framework for muscles to attach to the interior, allowing for powerful and efficient movement.

Epicuticle wax layer: Minimizes evaporation, enabling survival in dry environments.

Molting (ecdysis): Allows growth despite the rigid exoskeleton, though it temporarily makes them vulnerable.

➤ **Segmented Body & Jointed Appendages**

The name "arthropod" literally means "jointed foot." Their body plan is modular, consisting of repeating segments that have been specialized through evolution.

Tagmatization: Segments are fused into functional units called **tagmata** (such as the head, thorax, and abdomen). This allows for extreme specialization—the head for sensing/feeding, the thorax for locomotion, and the abdomen for reproduction/digestion.

Jointed Limbs: These appendages are highly adaptable. Depending on the species, they have been modified into legs for walking, wings for flying, mouthparts for piercing or chewing, antennae for sensing, and claws for defense, mandibles for chewing, pincers for defense.

➤ **Sensory & Nervous Systems**

Compound eyes: Many arthropods (like insects and crustaceans) have compound eyes consisting of thousands of individual visual units called **ommatidia**, which are excellent at detecting fast motion.

Antennae: These organs are packed with receptors for touch, vibration, and "smelling" chemical signals (pheromones).

Advanced nervous system: Supports complex behaviors like bee dances, camouflage, and social organization in ants and termites.

➤ **Ecological Versatility**

Habitat diversity: Arthropods colonize marine, freshwater, terrestrial, and aerial environments.

Feeding diversity: Herbivores, carnivores, omnivores, parasites, decomposers, filter feeders—occupying nearly every ecological niche.

Metamorphosis: Many arthropods, particularly insects, undergo **metamorphosis** (either complete or incomplete). This is a massive survival advantage because: **Niche Partitioning:** Larvae (like caterpillars) and adults (like butterflies) often eat different foods and live in different environments. This means the young do not compete with the adults for resources.

Flight: Insects access new food sources, escape predators, and disperse widely.

➤ **High Reproductive Potential**

Most arthropods have very short generation times and can produce a vast number of offspring in a single cycle. This allows their populations to recover quickly from environmental setbacks and enables rapid evolutionary adaptation to changes, such as developing resistance to pesticides.

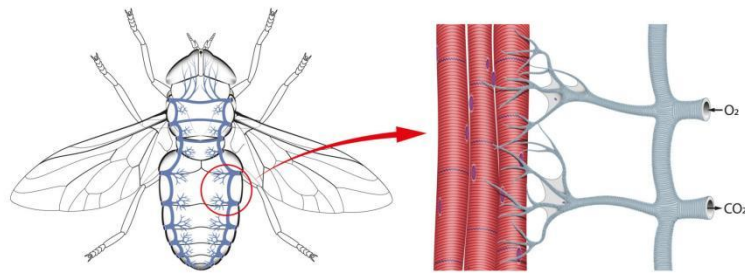
➤ **Efficient Respiratory Systems**

Arthropods have evolved respiratory systems that deliver oxygen directly to tissues, allowing for high metabolic rates:

Tracheal Systems: Insects use a network of tubes (tracheae) that open to the outside via pores called spiracles. This bypasses the need for a complex circulatory system to transport oxygen.

Gills and Book Lungs: Aquatic arthropods use specialized gills, while arachnids often use book lungs, which provide a large surface area for gas exchange.

INSECT TRACHEAL SYSTEM



Ecological Importance of Arthropods

Pollination: Insects like bees, butterflies, and beetles pollinate crops and wild plants, ensuring food production and biodiversity.

Decomposition: Termites, beetles, and crustaceans break down organic matter, recycling nutrients back into the soil.

Food chains: Arthropods serve as prey for birds, fish, amphibians, and mammals, forming a vital link in ecosystems.

Soil aeration: Burrowing arthropods (ants, termites) improve soil structure and fertility.

Economic Importance

Agriculture: Bees enhance crop yields through pollination; silkworms produce silk; lac insects produce shellac.

Fisheries: Crustaceans like crabs, lobsters, and shrimps are major sources of food and income.

Biological control: Predatory insects (ladybirds, wasps) help control agricultural pests, reducing reliance on pesticides.

Medical & Health Importance

Disease vectors: Mosquitoes transmit malaria, dengue, and yellow fever; tsetse flies spread sleeping sickness.

Research models: Fruit flies (*Drosophila*) are widely used in genetics and developmental biology studies.

Pharmaceuticals: Some arthropod venoms and secretions are being studied for medical applications.